CHEMISTRY - STANDARD 9

Chapter: 1

Structure of Atom

1. On what basis elements are arranged in Mendeleeve's periodic table?
a- atomic number
b- number of neutrons
c- atomic mass
d- none of these.

2. Who revised Mendeleeve's periodic law?
a- Neils Bohr
b- Newlands
c- Henry Mosely
d- Rutherford.

3. Elements in the same group of the periodic table tend to have :
a- same number of neutrons
b- similar Chemical properties
c- same number of electrons
d- All of the above

4. Name of the scientist who proposed Modern periodic law is----------
a) Mendeleeve
b) Moseley
c) J.J Thomson
d) Rutherford.

5. Anode rays are streams of:
a) Electrons
b) Positively charged particles
c) Neutrons
d) Negatively charged particles

6. Anode rays are formed when
a) a perforated cathode is used
b) a cathode having no perforation is used
c) gas filled in the discharge tube is removed completely

7. The discharge tube must be filled with _____ gas to generate protons

8. The presence of positive charge in the anode rays was first identified by_____

9. Explain the activities that led to the discovery of the proton.

10. J.J. Thomson's model of the atom is called the "plum pudding model" because:
a) Electrons are scattered throughout like plums in a pudding.
b) The model is positively charged.
c) The model is negatively charged

11. What did Thomson's experiment with cathode ray tube help him discover?
a) Protons
b) Neutrons
c) Electrons
d) Positive ions

12. According to Thomson model, an atom is electrically _____ because the positive and negative charges balance each other.

13. What are the two limitations of Thomson's plum pudding model?

14. In Rutherford's atomic model, the positively charged particles are concentrated in a region called the:
a) Nucleus
b) empty space

15. Rutherford's gold-foil experiment involved bombarding a thin sheet of:
a) Silver
b) Gold
c) Copper
d) Aluminium

16. Assertion: Rutherford's model has limitations and eventually new models were accepted.
Reason: The model could explain the stability of the atom.
(i) Both Assertion (A) and Reason (R) are true and R is the correct explanation of A.
(ii) Both (A) and (R) are true but R is not the correct explanation of A.
(iii) (A) is true but (R) is false.
(iv) (A) is false but (R) is true.

17. How did the alpha particles deflect in the Rutherford scattering experiment? Match the columns A and B suitably.

A	B
Most of the Alpha particles	Deflected by 180 Degree
Some of the alpha particles	Passed through the foil without deflection
A few alpha particles.	Deflected through a small angle.

18. Match the following terms with their descriptions:

A	B
a. Nucleus	i) Positively charged particles used in the experiment
b. Alpha particles	iii) The bending of the path of a particle
c Deflection.	ii) The tiny, dense center of the atom

19. According to Rutherford's model, what is the name of the positively charged center of the atom?

20. Where is the positive charge and most of the mass of an atom concentrated according to the model?

21. Explain the demerit of Rutherford's model of atom.

22. Neutrons are found in the: a) Nucleus of the atom b) orbits c) Both a) and b) d) None of the above

23. What is the charge of a neutron?
a) Positive
b) Negative
c) No charge
d) None of the above

24. The element Aluminium has 13 protons. But its mass is 27. Find out the number of neutrons in the nucleus.

25. Explain the role of neutrons in determining the mass number of an atom.

26. The path taken by an electron around the nucleus is called
a) Circular path
b) Revolving path
c) Orbits

27. The reason why orbits are called energy levels
a) Since all orbits have equal energy
b) Since each orbit has fixed energy
c) Since electrons are in orbits

28. What happens to the energy of an electron when it moves from a higher energy level to a lower energy level?
a) Released
b) absorbed
c) No difference

29. What happens to the energy of the orbit as the distance from the nucleus increases?
a) Increases
b) decreases
c) No difference

30. The scientist who proposed the concept of orbit
a) J J Thompson
b) Niels Bohr
c) Chadwick

31. Which of the following is not a particle in the nucleus?
a) Electron
b) Neutron
c) Proton

32. Which of the given particles has the least mass?
a) Electron
b) Neutron
c) Proton

33. Which of the following determines whether an atom belongs to a particular element.?
a) Number of electrons
b) Number of protons
c) Number of neutrons
d) Mass number

34. An element has 12 protons in its nucleus. What is its atomic number?
a) 10
b) 12
c) 24
d) None of these

35. If Z denotes the Atomic number and A denotes the mass number of an atom, which of the following represents the number of neutrons?
a) Z- A
b) A-Z
c) A+Z
d) Z+A

36. Explain the difference between atomic number and mass number.

37. The mass number of the element Titanium is 48. It has 22 electrons.
a) Find the number of protons and neutrons.
b) Represent the element Titanium (Ti) using the symbol.

38. The atomic number of an element is 30. There are 35 neutrons present in its nucleus.
a) What is its mass number?
b) How many electrons are present in it?

39. Assertion (A) : Atoms of the same element can have different masses.
Reason (R) : Elements have a fixed number of neutrons.
(a) A and R are both true and R is the correct explanation of A.
(b) A and R are both true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

40. Assertion (A) : The number of protons in an atom determines its chemical behavior.
Reason (R) : Protons are located in the nucleus
a) A and R are both true and R is the correct explanation of A.
(b) A and R are both true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

41. The maximum number of electrons that can be accommodated in the second orbit (L shell) is------
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42. Which of the following orbit has the lowest energy ?
a) N
b) M
c) K
d) L

43. Which is the correct electronic configuration of sodium (11 Na)
a) 2,9
b) 2,7,2
c) 2,8,1
d) none of these

44. Which of the following elements has a completely filled outermost shell?
(a) Hydrogen (H)
(b) Helium (He)
(c) Lithium (Li)
(d) Beryllium (Be)

45. The maximum number of electrons that can be accommodated in the outermost orbit of any atom is .....

46. The electronic configuration of an element X is 2,8,7.
a) What is its atomic number?
b) How many electrons are present in its M shell?
c) Find the number of electrons in the orbit with the lowest energy.

47. Write the electronic configuration for the following elements:
(a) Nitrogen(N) (Atomic number =7)
(b) Aluminium (Al) (Atomic number =13)
(c) Argon (Ar) (Atomic number = 18)

48. Assertion (A) : Neon has a completely filled outermost orbit (shell)
Reason (R) : The electronic configuration of Neon (Ne) is 2,8.
(a) Both A and R are true and R is the correct explanation of A.
(b) A is true but R is not the correct explanation of A.
(c) A is false but R is true.
(d) A is true but R is false

49. Which isotope of hydrogen does not contain neutrons?

50. Which isotope is used for the study and treatment of the thyroid gland?

51. Which isotope is used in cancer treatment?

52. Which of the given statements is correct?
a) Isobars have the same atomic number and different mass numbers.
b) Isobars have the same number of protons.
c) Isobars have the same mass number and different atomic numbers.